Azad Singh, PhD

Curriculum Vitae

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Professional Summary

Researcher in **Multimodal Machine Learning**, **Vision-Language Models**, and **Representation Learning**, with deep expertise in self-supervised and discrete representation learning (codebook/vector quantization) that bridges continuous visual features with structured symbolic spaces—a core building block for cross-modal alignment. Published in IEEE TNNLS (IF: 10.2), IEEE JBHI (IF: 7.7), MICCAI, WACV, ECAI, and NeurIPS workshop. Currently building **LLM-driven foundation models** for text-prompt-based 3D medical segmentation, integrating large language models with vision encoders for zero-shot cross-modal understanding. Seeking to advance multimodal foundation models across medical and general domains.

Research Interests

Multimodal Learning, Vision-Language Models, Foundation Models, Representation Learning, Self-Supervised Learning, Cross-Modal Alignment, Medical Image Analysis, Computer Vision.

Experience

Feb 2026 – Present **PhD Research Intern – Multimodal AI for 3D Medical Segmentation**, *Microsoft Research*, Bangalore, India.

- Building foundation models for **text-prompt-based anatomical and pathological segmentation** in 3D CT/MRI volumes
- Integrating large language models (LLMs) with 3D vision encoders for **cross-modal understanding** from natural language descriptions
- Developing zero-shot and few-shot segmentation pipelines driven by clinical language prompts
- Extending self-supervised discrete representation techniques to 3D volumetric data

Aug 2025 – Feb 2026 **Assistant Professor**, *School of Computer Science & Engineering, Bennett University*, Greater Noida, India.

- Taught undergraduate courses in Data Structures, Algorithms, and Machine Learning
- Designed curriculum materials integrating research-led pedagogy with industry-relevant skills

2017 – 2019 **Assistant Professor (Contract)**, *National Institute of Technology*, Hamirpur, H.P., India.

- Delivered core CS courses across UG/PG programmes for 2 years

Technical Skills

Programming Languages: Python (advanced), C/C++

Deep Learning Frameworks: PyTorch, TensorFlow, Keras

Multimodal & LLM Tools: Hugging Face Transformers, LLaMA, CLIP, Segment Anything (SAM)

3D Medical Imaging: MONAI, nibabel, SimpleITK

Libraries & Tools: OpenCV, Scikit-learn, Git, NumPy, Pandas, Weights & Biases

Domain Expertise: Multimodal Learning, Vision-Language Models, Cross-Modal Alignment, Self-Supervised Learning, Representation Learning, Medical Image Analysis

Research Highlights

Discrete Representations for Cross-Modal Alignment Pioneered **codebook/vector quantization-based SSL** methods (CoBooM – MICCAI 2024; DiSSECT – IEEE JBHI, under review; ECAI 2025 Oral) that learn structured discrete visual dictionaries from unlabelled images. These discrete bottlenecks bridge continuous visual features with symbolic representations—the same interface needed for aligning images with language in VLMs.

Multimodal Fusion & Survival Prediction Co-developed **RobSurv** (under review, Computer Methods and Programs in Biomedicine), a vector quantization-based multimodal learning framework that fuses histopathology images, genomic data, and clinical metadata for robust cancer survival prediction—demonstrating cross-modal alignment at scale.

Cross-Domain Temporal Representation Developed a **multi-frequency bootstrapping** framework (IEEE TNNLS 2025, IF: 10.2) for time-series representation learning, demonstrating that self-supervised bootstrapping principles generalize across imaging and temporal signal modalities (ECG, EEG, sensor data)—a key step toward multimodal signal integration.

Dense Semantic Alignment Proposed **OTCXR** (WACV 2025), reformulating SSL alignment as an optimal transport problem for dense semantic invariance—a principled approach to fine-grained feature correspondence directly applicable to vision-language matching.

Embedding Regularization at Scale Introduced multi-level variance-covariance regularization (**MLViCX**, IEEE JBHI 2024, IF: 7.7) preventing representational collapse while preserving both global context and fine-grained detail across 4 public benchmarks with >3% gains over SOTA.

Education

2019–2026 **PhD, Computer Science & Engineering, Indian Institute of Technology, Jodhpur**,
Self-supervised Encoding of Chest Radiographs for Learning Generalized Representations .

2014–2016 **M.Tech, Computer Science & Engineering, Kurukshetra University**,
Digital Forensics, Memory Forensics Percentage: 71.8%.

2009–2013 **B.Tech, Computer Science & Engineering, Kurukshetra University,** Percentage: 70.0%.

Publications

Journal Articles

2025 **Azad Singh** and Deepak Mishra. Self-supervised contextual representations of chest x-ray images. *IEEE Signal Processing Letters*, 2025. **IF: 3.9**.

2025 **Azad Singh**, Vandan Gorade, and Deepak Mishra. Large scale time-series representation learning via simultaneous low- and high-frequency feature bootstrapping. *IEEE Transactions on Neural Networks and Learning Systems*, 2025. **IF: 10.2**.

2024 **Azad Singh**, Vandan Gorade, and Deepak Mishra. Mlvicx: Multi-level variance-covariance exploration for chest x-ray self-supervised representation learning. *IEEE Journal of Biomedical and Health Informatics*, 2024. **IF: 7.7**.

Communicated Journal Articles

2024 **Azad Singh**, and Deepak Mishra, SiMEX: Siamese Ensembling for Multi-Latent Exploration in Self-Supervised Chest X-ray Representation Learning, In *Computers in Biology and Medicine, Elsevier* **IF:7.7**.

2025 **Azad Singh**, and Deepak Mishra, DiSSECT: Structuring Transfer-Ready Medical Image Representations through Discrete Self-Supervision, In *IEEE Journal of Biomedical and Health Informatics (JBHI)* **IF:7.7** .

2025 Aiman Farooq, ***Azad Singh***, *Deepak Mishra and Santanu Chaudhury*, RobSurv: Multi-Modal Learning for Robust Cancer Survival Prediction, In *Computer Methods and Programs in Biomedicine***IF:4.8** .

In Conference Proceedings and Workshops

2025 **Azad Singh** and Deepak Mishra. Scale-aware adaptive feature quantization for robust medical image representation learning. In ***ECAI***, 2025. CORE **A**.

2025 **Azad Singh**, Wilfred Kisku, Amandeep Kaur, and Deepak Mishra. Vit coupled efficient cmos image sensor with sparse acquisition and patch selection. In ***ISCAS***, 2025. Qualis **A1**.

2025 **Azad Singh**, Vandan Gorade, and Deepak Mishra. Otcxr: Rethinking self-supervised alignment using optimal transport for chest x-ray analysis. In ***WACV***, 2025. CORE **A**.

2025 Gyandera Singh, **Azad Singh**, Aiman Farooq, and Deepak Mishra. Ribcageimp: A deep learning framework for 3d ribcage implant generation. In ***ICVGIP.*** , 2025.

2025 Gajendra Singh, Aiman Farooq, **Azad Singh**, Deepak Mishra, Rahul Chaudhury, and Pushpinder Singh Khera. Guide2heart: Proximity guidance for standard echocardiographic view. In ***Simplifying Medical Ultrasound, ASMUS, MICCAI Workshop***, 2025. CORE **A**.

2024 **Azad Singh** and Deepak Mishra. Idqce: Instance discrimination learning through quantized contextual embeddings for medical images. In ***ICPR***, 2024. CORE **B**.

2024 **Azad Singh** and Deepak Mishra. Coboom: Codebook guided bootstrapping for medical image representation learning. In ***MICCAI***, 2024. CORE **A**.

2022 **Azad Singh** and Deepak Mishra. Branching out for better byol. In *NeurIPS Workshop: Self-Supervised Learning - Theory*, 2022. CORE **A***.

2022 Vandan Gorade, **Azad Singh**, and Deepak Mishra. Mbgrlp: Multiscale bootstrap graph representation learning on pointcloud. In *Student Abstract, AAAI*, 2022. CORE **A***.

Fellowships & Awards

2025 **Oral Presentation** – Scale-Aware Adaptive Feature Quantization for Robust Medical Image Representation Learning, *ECAI 2025*, Bologna, Italy, October 27–30, 2025

2025 **Best Poster (Runner-Up)** – Guide2Heart: Proximity Guidance for Standard Echocardiographic View, *ASMUS Workshop, MICCAI 2025*, Daejeon, Republic of Korea, September 27, 2025

2025 **Invited Presentation** – *ConfAI 2025*, Plaksha University, Mohali, November 28–30, 2025

2025 **Poster Presentation** – Graduate Forum, *IndoML 2025*, BITS Pilani – Hyderabad Campus

2024 ACM India–IARCS Travel Grant

2024 RISE–MICCAI & Anusandhan National Research Foundation (ANRF) Travel Support

2021 Best Teaching Assistant Award – Data Structure & Algorithms

2019–2023 MHRD Scholarship – Doctoral Fellowship for PhD Research

Referees

Dr. Deepak Mishra
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